

National University of Computer and Emerging Sciences, Lahore Campus



Course:	Computer Organization and Assembly Language	Course Code: EE2003
Program:	BS (CS, DS)	Semester: Fall 2021
Duration:	60 Minutes	Total Marks: 30
Paper Date:	2-Nov-2021	Weightage: 15
Section(s):	All	Page(s): 8
Exam:	Midterm II	Section: _____
		Roll No: _____

Instruction/Notes:

- Exam is Open book, Open notes.
- Properly comment your code.
- You **CANNOT** use an instruction **NOT** taught in class.
- If there is any ambiguity, make a reasonable assumption. Questions during the exam are not allowed.
- Write your answer in the space provided. You **can take extra sheets BUT they WON'T BE ATTACHED WITH THE QUESTION PAPER OR MARKED.**
- All other rules pertaining to examinations as per NUCES policy apply.

Question 1 [15 Marks]: Short Questions

- i. [2 marks] Consider a subroutine TempSBR that uses the stack to return three output values (*each of size 1 word*) through the stack. Write a statement that will create the space for these three output variables before calling this TempSBR.

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- ii. **[6 marks]** Consider the following subroutine, which calculates the factorial of a number (*size = 1 word*) placed at the stack as a parameter and outputs the answer on the stack (*size = 1 word*). However, the code has some logical errors. Correct those errors so that the required functionality can be achieved. You can ADD or MODIFY existing lines, but you cannot REMOVE any line.

<pre>factorial: push bp mov bp, sp push ax push bx push dx mov ax, [bp+8]; copying the input cmp ax, 0 ja L1 mov word [bp+10], 1; returning the result jmp L2 L1: sub sp, 2 dec bp push bp; passing parameter for recursive subroutine call factorial; recursive subroutine call returnFact: pop bx mov dx, 0 inc ax mul bx mov [bp+10], ax; returning the result L2: pop dx pop bx pop ax pop bp ret 6</pre>	; Rewrite your code here
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- iii. **[3 Marks]** Consider the code given below, write out the sequence in which the instructions are executed. Each executable instruction in code is numbered so your answer should be as follows:

Sample answer:

Instructions executed in following order

I11

I6

I10

....

You also have to briefly explain the working of this program.

	<pre>[org 0x0100] I1 jmp start my_rout: I2 mov ax, 0x8434 I3 mov bl, 0x85 I4 div bl I5 mov ax, 0xffff I6 mov dx, 0x0100 I7 mov bl, 0x3 I8 div bl I9 ret start: I10 call my_rout I11 mov ax, 0x4c00 I12 int 0x21</pre>	<u>Solution:</u>
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- iv. **[4 Marks]** In the code given below, we are copying the data of video memory from one location to another using string instructions. As a result of the execution of this code, what will be the changes on the screen?

<pre>[org 0x0100] jmp start movepixels: push ax push bx push cx push si push di push es push ds mov ax, 0xb800 mov es, ax mov ds, ax mov si, 0 mov di, 80 mov bx, 0 ; (code is continued in the second column)</pre>	<pre>loop1: mov cx, 80 cld rep movsb add si, 80 add di, 80 add bx, 1 cmp bx, 25 jne loop1 pop ds pop es pop di pop si pop cx pop bx pop ax ret start: call movepixels mov ax, 0x4c00 int 0x21</pre>
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Solution:

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Question 2 [15 Marks]: Draw a triangle with two given points i.e. A (x1, y1) and B (x2, y2).

i. **[3 Marks]** Triangle must be isosceles (two sides equal) and right (one 90-degree angle), for that purpose check two conditions given below.

a) y1 must be less than y2 and x1 must be less than x2.

b) (x2-x1) must be equal to (y2-y1).

No need to check other conditions as these two conditions are enough.

ii. **[2 Marks]** Clear screen with white background.

iii. **[7 Marks]** Only print the boundary of the triangle with red color and asterisk character (ASCII= 2A-Hex,42-Decimal).

Hint: Write a generic subroutine to print an asterisk on a single point. Use loops to print borders.

iv. **[3 Marks]** Write a program with proper subroutine names and stack implementation is compulsory for parameter passing.

Note: You can't use software interrupts. You should use hard code inputs but functions should be generic. It should run properly on any inputs.

Example 1:

Input: A (7, 8) and B (10, 11)

Output : (7,8)

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      *
      *      *
      *          *
      *      *      *      *
                                     (10,11)
```

Example 2:

Input: A (10, 11) and B (7, 8)

Output: No printing on screen

Example 3:

Input: A (7, 8) and B (10, 10)

Output: No printing on screen

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Write your code below

